

Assignment-3: Exploratory Data Analysis (EDA) and Hypothesis Testing on Preprocessed Earthquake Dataset

Objective

Conduct comprehensive exploratory data analysis (EDA) and statistical hypothesis testing on the preprocessed earthquake dataset to uncover patterns, relationships, and significant differences in seismic characteristics that influence earthquake damage potential and geographical distribution.

Dataset

Preprocessed Earthquake Dataset from `data_preprocessing_v3.ipynb`

- **Shape:** 109,848 rows × 17 columns
 - **Key Features:** latitude, longitude, depth, magnitude (mag), moment magnitude (Mw), rms, magType, type, status, temporal features (Year, Month, Day, Hour, Minute, Second, DayOfWeek), damage_potential
 - **Target Variable:** damage_potential (0-10 scale using HAZUS methodology)
 - Link: [Preprocessed Dataset](#)
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Part 1: Exploratory Data Analysis (EDA)

EDA Task 1: Temporal Distribution Analysis

Objective: Analyze earthquake frequency and magnitude trends over time.

Tasks:

- Plot earthquake frequency by Year, Month, and DayOfWeek
- Identify seasonal patterns and temporal trends
- Calculate mean magnitude and damage potential by temporal dimensions
- Visualize time series of earthquake counts and average magnitude over years
- Create heatmap of earthquakes by Month and DayOfWeek

Deliverable: Line plots, bar charts, heatmaps, and summary statistics documenting temporal patterns.

EDA Task 2: Geographical Distribution Analysis

Objective: Explore spatial patterns of earthquake epicenters.

Tasks:

- Generate scatter plot of latitude vs. longitude colored by magnitude or damage_potential
- Create density maps or 2D histograms showing earthquake hotspots
- Analyze earthquake distribution by geographical regions (latitude/longitude bins)
- Calculate correlation between geographical coordinates and seismic characteristics
- Identify high-risk geographical areas based on damage_potential

Deliverable: Scatter plots, density maps, geospatial visualizations, and regional summary statistics.

EDA Task 3: Magnitude and Damage Potential Relationship

Objective: Analyze the relationship between earthquake magnitude and damage potential.

Tasks:

- Create scatter plot: Magnitude (mag) vs. Damage Potential with depth color-coding
- Plot Moment Magnitude (Mw) vs. Damage Potential
- Generate distribution histograms for mag, Mw, and damage_potential (with KDE)
- Calculate correlation coefficients (Pearson and Spearman) between mag/Mw and damage_potential
- Analyze outliers and their characteristics

Deliverable: Scatter plots, histograms, KDE plots, and correlation matrices.

EDA Task 4: Depth Analysis and Its Impact

Objective: Investigate how depth influences seismic characteristics and damage.

Tasks:

- Create box plots: Depth vs. Magnitude (stratified by depth ranges: shallow, intermediate, deep)
- Analyze depth distribution with histogram and statistics (mean, median, std)
- Plot depth vs. damage_potential using scatter and 2D density plots
- Calculate mean damage_potential for different depth ranges
- Identify whether shallow or deep earthquakes cause more damage

Deliverable: Box plots, histograms, scatter plots, and depth-based summary tables.

EDA Task 5: Categorical Feature Analysis

Objective: Explore relationships between earthquake types, magnitude types, and status.

Tasks:

- Analyze value counts for magType, type, and status columns
- Create bar charts showing frequency distribution of each categorical variable
- Generate grouped bar charts: magType vs. mean magnitude and damage_potential
- Create cross-tabulation (contingency tables) between categorical variables
- Visualize earthquake type distribution and its association with damage_potential
- Calculate Cramér's V statistics for categorical associations

Deliverable: Bar charts, contingency tables, Cramér's V matrices, and frequency distributions.

Part 2: Hypothesis Testing

Hypothesis 1: Magnitude Difference by Earthquake Type

Research Question: Do different earthquake types (e.g., earthquake, explosion, quarry blast) have significantly different magnitudes?

Hypotheses:

- H0: The mean magnitude across all earthquake types is equal.
- H1: At least one earthquake type has a significantly different mean magnitude.

Statistical Test: One-Way ANOVA or Kruskal-Wallis H-test (if normality violated)

Execution Steps:

1. Group data by 'type' (earthquake, explosion, quarry blast, etc.)
2. Calculate mean and variance of 'mag' for each group
3. Perform ANOVA or Kruskal-Wallis test
4. Report F-statistic (or H-statistic), p-value, and effect size

Interpretation: If p-value < 0.05, reject H0 and conclude earthquake types differ significantly in magnitude.

Hypothesis 2: Damage Potential by Magnitude Categories

Research Question: Is there a significant difference in damage_potential between high-magnitude and low-magnitude earthquakes?

Hypotheses:

- H_0 : The mean damage_potential of high-magnitude earthquakes ($M_w \geq \text{median}$) equals that of low-magnitude earthquakes ($M_w < \text{median}$).
- H_1 : The means differ significantly between magnitude categories.

Statistical Test: Independent samples t-test or Mann-Whitney U test (non-parametric alternative)

Execution Steps:

1. Partition data into two groups based on median M_w
2. Compare damage_potential distributions
3. Perform t-test or Mann-Whitney U test
4. Calculate p-value and 95% confidence intervals

Interpretation: If p-value < 0.05, reject H_0 ; high-magnitude earthquakes cause significantly different damage levels.

Hypothesis 3: Depth Impact on Magnitude

Research Question: Does earthquake depth significantly correlate with magnitude?

Hypotheses:

- H_0 : There is no significant correlation between depth and magnitude ($\rho = 0$).
- H_1 : There is a significant correlation between depth and magnitude ($\rho \neq 0$).

Statistical Test: Pearson Correlation Test or Spearman Rank Correlation (for non-linear relationships)

Execution Steps:

1. Calculate Pearson or Spearman correlation coefficient between depth and mag
2. Compute p-value and 95% confidence interval
3. Create scatter plot with regression line overlay
4. Assess correlation strength and practical significance

Interpretation: If p-value < 0.05 and $|\rho| > 0.3$, conclude significant correlation; deeper or shallower earthquakes have systematically different magnitudes.

Hypothesis 4: Temporal Seasonality in Earthquake Frequency

Research Question: Does earthquake frequency differ significantly across months (seasonal effect)?

Hypotheses:

- H0: Earthquake frequency is uniform across all months (no seasonal effect).
- H1: Earthquake frequency varies significantly by month.

Statistical Test: Chi-Square Goodness-of-Fit Test or One-Way ANOVA on monthly counts

Execution Steps:

1. Count earthquakes per month
2. Perform chi-square test against uniform distribution
3. Calculate chi-square statistic, degrees of freedom, and p-value
4. Visualize monthly distribution with confidence bands

Interpretation: If p-value < 0.05, reject H0; earthquakes show significant seasonal clustering.

Hypothesis 5: RMS Variability by Magnitude Type

Research Question: Do different magnitude type classifications (magType) show significantly different RMS (residual mean square) variability?

Hypotheses:

- H0: The mean RMS across all magType categories is equal.
- H1: At least one magType category has a significantly different mean RMS.

Statistical Test: One-Way ANOVA or Levene's Test (for homogeneity of variances)

Execution Steps:

1. Group data by magType
2. Calculate mean RMS for each group
3. Perform ANOVA and Levene's test for variance equality
4. Conduct post-hoc pairwise comparisons (Tukey HSD) if ANOVA is significant
5. Report F-statistic, p-value, and effect size (η^2)

Interpretation: If p-value < 0.05, reject H0; magnitude type classifications differ in RMS characteristics, suggesting data quality or measurement differences.

Deliverables

Jupyter Notebook Requirements:

1. **Section 1-5:** Detailed EDA implementation with:

- Clean, reproducible code for each EDA task
- High-quality visualizations (plots, heatmaps, geographical maps)
- Summary statistics and insights for each analysis
- Code comments explaining methodology

2. **Section 6-10:** Comprehensive hypothesis testing with:

- Complete implementation of all 5 hypothesis tests
- Statistical test outputs (test statistics, p-values, confidence intervals)
- Visualizations supporting test assumptions (normality plots, Q-Q plots, histograms)
- Clear interpretation of results in context
- Practical implications for earthquake risk assessment

3. **Final Section:** Integrated Report including:

- Summary of key EDA findings
- Hypothesis testing conclusions
- Recommendations for earthquake damage prediction modeling
- Limitations and future work suggestions

Output Requirements:

- Save notebook with all outputs (`.ipynb` with executed cells)
 - Export key visualizations as high-resolution images
 - Generate a summary CSV containing all statistical test results
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